Lecture plan · for the lecturer

Attention and the transformer

University, 4

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Score, then mix
- Add the original back

Activity: Two labs, one claim

1. Attention lab (10 min): One phone between two: tap “it”, point at the word that gets most of the attention, and say what was scored.
2. Block (10 min): Draw one transformer block on the board: attention mixes the items, the feed-forward layer works on each item on its own, and each adds its result back onto the original (the residual connection).
3. Eight notes lab (10 min): Before anyone presses the button, pairs say which pitch may follow E. Then they read the counts and the four notes it writes.
4. Mark (10 min): Each student writes what attention mixed and what the melody model counted. Mark the sentence that separates mixing from understanding.
5. Exit (5 min): The exit ticket: attention replaces which hope?

Discussion

- What was not counted?

Exit ticket

- Attention replaces which hope?

Try it in your next lecture
Put the attention lab and the Eight notes lab side by side on one slide and ask which one looks back.

Seminar sheet · for students

Attention and the transformer

University, 4

Name _____ Group _____ Date _____

1 **The last word of a sentence depends on its first word, 30 words back. Why does attention keep that link when the simple recurrent net loses it?**

- ◯ It was trained for longer
- ◯ It scores the first word directly, however far back it is
- ◯ It reads the sentence one word at a time

2 **The Eight notes lab composed a piece of music.**

- ◯ True
- ◯ False

3 **Match the piece.**

Query	• • • •	The label each item offers, matched against the query
Key	• • • •	Adding the original back
Residual connection	• • • •	What this item is looking for

4 **Order an attention step.**

- ☐ Mix the vectors with those weights
- ☐ Score every item
- ☐ Turn scores into weights that add to 1

Answers · for the lecturer

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1 It scores the first word directly, however far back it is

Attention compares the last word with every earlier word in one step, so distance does not weaken the link.

2 False

It continued the counts from one short tune.

3 Query — What this item is looking for
Key — The label each item offers, matched against the query
Residual connection — Adding the original back

The query is matched against every key to score the items.

4 Top to bottom: 3, 1, 2

The mix is last. The weights have to exist first.

For you to think about

Which claim will you not accept on a project poster?